

Design and Construction of a DC Voltage Supplier with Seven-Segment Display

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Abstract—This paper presents the design and construction of a variable DC power supply integrated with a seven-segment digital display. A variable DC power supply is a device that delivers electrical power to a load and allows the user to vary the output voltage within a specified range. Such a unit is among the most frequently used instruments in electronic workshops and laboratories, where a fluctuation-free, adjustable output is required. The proposed circuit is built around an adjustable three-terminal variable-voltage regulator LM338, a fixed 5 V regulator LM7805, and an LM337 negative regulator used for current regulation. The 220 V AC mains supply is stepped down by a transformer to a secondary output of $-12\text{ V}-0-+12\text{ V}$ AC at 3 A. A full-wave bridge rectifier converts this to pulsating DC, which is smoothed by filter capacitors to eliminate ripple. Both the positive and negative half-cycles are used to obtain positive and negative DC outputs simultaneously. The digital display section employs the CMOS analog-to-digital converter IC ICL7107CPLZ to measure and display the regulated output voltage and current on seven-segment displays.

Index Terms—DC power supply, seven-segment display, voltage regulator, LM338, LM337, ICL7107, rectifier, digital voltmeter

I. INTRODUCTION

A. Background

This work reports on a project carried out as part of the undergraduate electrical and electronics engineering curriculum. If electrical systems can convert AC power close to the point of use, the converters used for this purpose can be made more efficient and cost-effective. The central aim of the project is to convert the AC mains voltage into a variable, regulated DC voltage. The project therefore focuses on the design and construction of such a DC power-supply unit, with the measured output presented directly on a digital display.

B. Objectives

The objectives of the project are to:

- become familiar with the practical use of common electrical and electronic equipment;
- develop teamwork and collaborative design skills;
- transform a high-voltage AC input into a regulated, low-voltage DC output;
- increase flexibility of use by providing both dual fixed and variable voltage outputs; and
- develop a device that is safe, efficient and economical.

C. Working Mechanism

The device accepts a 220 V AC input, which is first reduced by a step-down transformer. The stepped-down voltage is then rectified by diodes so that current flows in only one direction. A filter capacitor smooths the pulsating rectified waveform into a nominally constant DC voltage. This smoothed DC voltage is subsequently applied to fixed and variable regulators to obtain the fixed and variable outputs, while a separate current-regulation stage limits the output current. The resulting output voltage and current are displayed visually using seven-segment indicators.

II. SYSTEM COMPONENTS

A high-quality power supply of this type is an electronic device whose output can be continuously varied between approximately zero and 20 V DC. An additional electronic current limiter restricts the output current from a few milliamperes (2 mA) up to a maximum of 3 A, which protects a circuit under test from damage. Visual indication of the supplied voltage and current is also provided. The hardware building blocks used to realize this system are described below.

A. Transformer

A transformer steps up or steps down an AC input voltage. The transformer used in this project has a single-primary, two-secondary (1P2S) configuration rated at 12 V-0-12 V, 3 A, and is used to step down the 220 V AC mains to 12-0-12 V.

B. Rectifier

A rectifier converts alternating current (AC), which periodically reverses direction, into direct current (DC), which flows in only one direction. A bridge rectifier built from 1N4007 diodes is used in this design.

C. Filter

Filters are electronic circuits used to remove unwanted frequency components from a signal. The filter stage in this design uses electrolytic capacitors, specifically a 4700 μF capacitor, whose selection is derived analytically in Section IV.

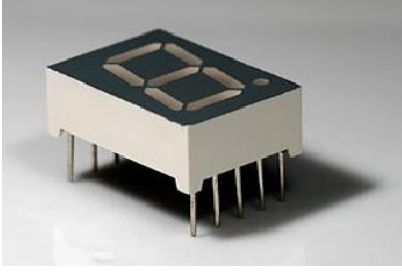


Fig. 1. Seven-segment display module used for the voltage and current readout.

D. Seven-Segment Display

A seven-segment display is a set of seven bar-shaped light-emitting-diode (LED) elements arranged to form a squared-off figure-8, used to present numerical information. Seven-segment displays are widely used in digital clocks, electronic meters, calculators, and other devices that must display numerical data, as shown in Fig. 1.

E. Voltage Regulators

A regulator is a circuit that maintains a fixed or constant output voltage, fixing the output to a desired level and holding that value despite variations at its input or output. In this design, the IC LM338 provides the variable positive voltage regulation, the IC LM7805 provides the fixed 5 V regulation, and the IC LM337 is employed for current regulation.

F. A/D Converter and Display Driver

The ICL7107CPLZ analog-to-digital converter IC has a very high input impedance and requires no external display-drive circuitry. Its on-board active components include polarity and digit drivers, segment decoders, a voltage reference, and a clock circuit, and it can directly drive a common-anode LED display. It is used here for the visual indication of voltage and current. Its principal advantages are that no external display driver is required, it offers good measurement accuracy, and it is compact, fitting on a 35×50 mm board.

G. Simulation Tool and Thermal Management

Multisim, an electronic schematic-capture and simulation package, was used to simulate the circuit prior to its implementation on a breadboard and printed circuit board (PCB). A metallic heat sink is used to draw excess heat away from the regulator ICs and to prevent thermal overload.

III. SYSTEM DESIGN AND OPERATION

A. System Overview

The 220 V, 50 Hz AC mains supply is fed directly to the one-primary, two-secondary (1P2S) transformer, which steps it down to approximately 12-0-12 V. This voltage is rectified by a bridge rectifier connected immediately after the transformer. The rectified signal is a continuously pulsating DC waveform; filter capacitors then convert this pulsating DC into a smooth, non-pulsating DC supply, which is subsequently regulated by the ICs described above:

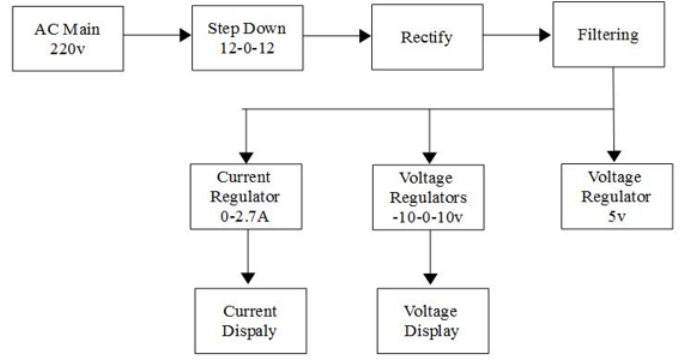


Fig. 2. Block diagram of the DC power supply system.

- **LM7805** regulates a constant 5 V output;
- **LM337** is a variable negative regulator capable of supplying in excess of 5 A over -1.2 to -32 V; and
- **LM338** is a variable positive regulator capable of supplying in excess of 5 A over 1.2 – 32 V, and is also used for current regulation.

Separate output pins for the positive voltage, the negative voltage, and a common ground are brought out from the circuit so that the desired combination of voltages can be selected. A 1.5 A fuse is placed in series with the transformer to protect the circuit against short-circuit damage, and a heat sink together with a cooling fan is used to prevent the regulator ICs from overheating. The output voltage and current are displayed simultaneously through the display section, which is built around the ICL7107CPLZ, a high-performance, low-power $3\frac{1}{2}$ -digit A/D converter that integrates seven-segment decoders, display drivers, a voltage reference and a clock, driving a common-anode seven-segment display. The overall system architecture is shown in Fig. 2.

B. Digital Display Panel

1) *Digital Voltmeter*: The digital voltmeter displays the measured DC output voltage of the supply as discrete numerals on the seven-segment display. The display section uses the CMOS IC ICL7107CPLZ, a typical A/D converter that measures voltage using the dual-slope technique. It is a complete integrated package comprising an integrator, a level comparator, a basic clock, counters, and a display driver, as illustrated in Fig. 3. The unknown input voltage V_x is fed to the integrator through a switch S for a fixed time interval T , which is set by the clock frequency of the decimal counter; during this interval the integrator capacitor charges at a rate proportional to V_x . At the end of interval T , the switch S is shifted to the reference voltage V_{ref} (100 mV), which is applied for a time t determined by counting clock pulses until the capacitor voltage returns to its reference level. From the similar-triangles relationship of the integrator ramp,

$$\frac{V_x}{T} = \frac{V_{ref}}{t} \implies V_x = \frac{T}{t} V_{ref}, \quad (1)$$

the count accumulated during t is proportional to V_x and is displayed as the measured voltage.

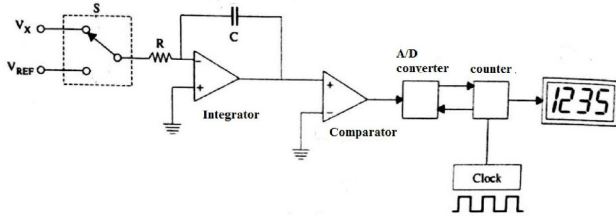


Fig. 3. Block diagram of the digital voltmeter based on the dual-slope A/D technique.

2) *Digital Ammeter*: The digital ammeter measures the DC output current of the supply in the same manner as the digital voltmeter, using the same ICL7107CPLZ IC with a slight modification of the passive components; its block diagram is essentially identical to that of the digital voltmeter shown in Fig. 3.

IV. DESIGN CALCULATIONS

A. Design of the Variable DC Circuit

The circuit is designed to produce both positive and negative output voltages using the two outputs of a bridge rectifier, one for the positive side and one for the negative side. The transformer steps the mains voltage down to 12 V (RMS), and the voltage drop across the two conducting diodes of the bridge is $0.7 \times 2 = 1.4$ V. The peak voltage is obtained by multiplying the RMS voltage by 1.414, since

$$V_{rms} = \frac{V_p}{1.414}. \quad (2)$$

Hence, the peak voltage after the bridge rectifier is

$$V_p = 12 \times 1.414 = 16.968 \text{ V}. \quad (3)$$

Subtracting the diode drop of 1.4 V from this peak value gives a usable peak DC voltage of 15.57 V. Because the rectified DC still contains significant ripple, a smoothing (filter) capacitor is required; the amount of ripple depends on the capacitance C , the DC voltage V , the load current I , and the AC line frequency f .

B. Selection of Filter Capacitance

For a target ripple factor $r = 0.167$ at a line frequency $f = 50$ Hz, the maximum load resistance is

$$R = \frac{V_{rated}}{I_{rated}} = \frac{12}{3} = 4 \Omega. \quad (4)$$

Using the ripple relation

$$r = \frac{1}{4\sqrt{3}fCR} \implies C = \frac{1}{4\sqrt{3}frR}, \quad (5)$$

substituting the above values gives

$$C = \frac{1}{4 \times 1.732 \times 50 \times 0.167 \times 4} = 4.3 \text{ mF}. \quad (6)$$

A standard capacitor of 4700 μF was therefore selected for the positive section, with an identical, independent 4700 μF capacitor used for the negative section.

At a 220 V AC mains input, the maximum rectified and smoothed voltage available at the regulator inputs is thus 15.57 V, which is supplied to the LM338 and LM337 regulator ICs. With $V_{in} = 15.57$ V, the maximum output is $V_{out} = 13$ V, giving a nominal adjustable range from 1.25 V to 13 V. A 1.5 V Zener diode is added to lower this limit, giving a theoretical output range of 0–11.5 V; in practice, the positive section delivers 0.4–10 V and the negative section delivers –0.2 to –10.1 V.

C. Voltage-Regulator Resistor Selection

For the voltage divider that programs the LM338/LM337 regulators, with $V_{out} = 10$ V and the internal reference $V_{ref} = 1.25$ V,

$$V_{out} = V_{ref} \left(1 + \frac{R_1}{R_2} \right), \quad (7)$$

which requires $R_1/R_2 = 7$. Consequently, the resistor ratio R_1/R_2 must be chosen greater than 7 to cover the full intended output range.

V. CIRCUIT IMPLEMENTATION

The complete power-supply circuit, comprising the transformer, bridge rectifier, filter capacitors, and the LM338/LM337/LM7805 regulator network, was implemented and simulated in Multisim before hardware construction, as shown in Fig. 4. Separate positive, negative, and common ground rails are brought out so that the fixed and variable outputs can be selected independently.

A dedicated current-regulator stage, built around the LM338 with a low-value sense resistor and a potentiometer for adjustable current limiting, is used to protect the circuit and the load under test; its schematic is shown in Fig. 5.

The A/D converter and display driver stage is built around the ICL7107CPLZ configured as shown in Fig. 6, using LED display components selected for a 200 mV full-scale reading; the reference and integrator components of this test circuit were scaled to match the voltage and current ranges of the power supply.

VI. RESULTS AND IMPLEMENTATION

The following tasks were completed during construction and testing of the unit:

- The step-down transformer was connected to the AC mains and stepped the input down to 12-0-12 V.
- The stepped-down, low voltage was rectified into pulsating DC using a bridge rectifier; each end of the rectifier produced a positive and a negative pulsating DC output.
- The pulsating DC from each rectifier output was smoothed using a 4700 μF electrolytic filter capacitor.
- Each filtered DC signal was fed to its corresponding regulator IC: the positive rail to the LM338 and LM7805, and the negative rail to the LM337.
- The LM338 delivered an output of +0.4 V to +10 V, the LM337 delivered –0.4 V to –10 V, and the LM7805 delivered a constant 5 V.

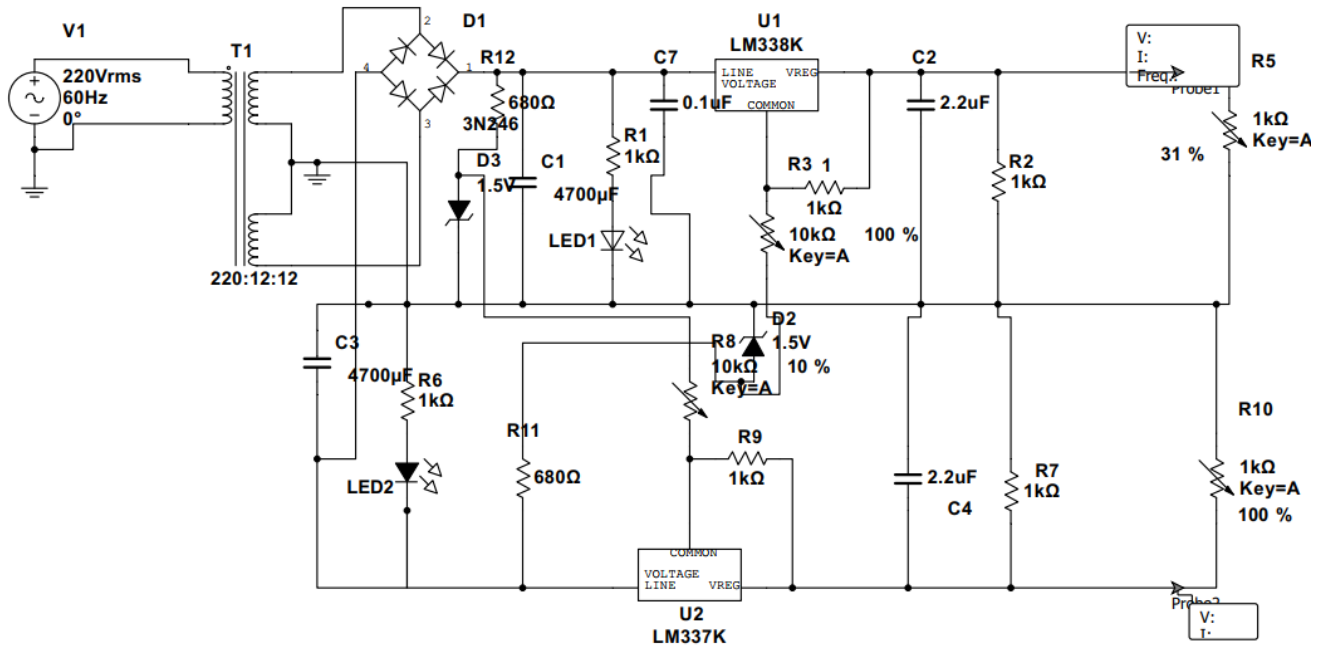


Fig. 4. Schematic of the dual DC voltage supplier circuit (0–20 V), designed and simulated in Multisim.

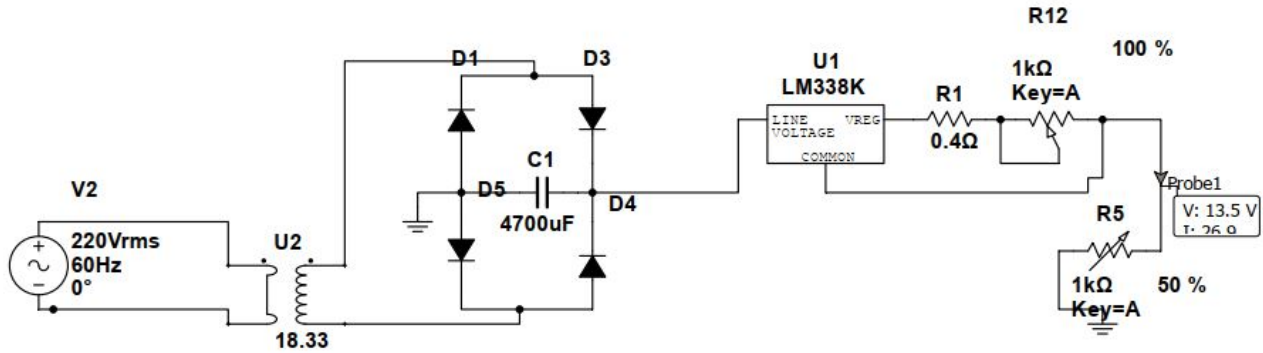


Fig. 5. Schematic of the adjustable current-regulator circuit built around the LM338.

- Overall, the completed system is able to deliver up to 20 V DC.
- The output voltage is displayed using the ICL7107CPLZ-based display section.
- PCB design, printing, and etching were completed, and the ICL7107 was tested and implemented for current measurement and display.
- The PCB was fabricated, and the complete unit was assembled into its enclosure.

The measured output values of the constructed unit were found to correspond closely to the values obtained analytically in Section IV, confirming the validity of the design approach.

VII. CONCLUSION

Most domestic and laboratory equipment of small power ratings must be supplied with electricity in one form or

another, and there is often a shortage of purpose-built power supplies suited to a given domestic, laboratory, or project application. To address this need, a multi-output DC power-supply unit with a digital display was designed, simulated, and constructed. The variable DC output produced values ranging from 0 to 20 V, while the regulated dual outputs produced 5 V and ± 10 V, respectively. Output measurements showed that the developed power supply performed effectively, with measured values showing only minor deviation from the nominal design values. The resulting system is inexpensive, robust, and well suited to domestic, laboratory, and project-development applications.

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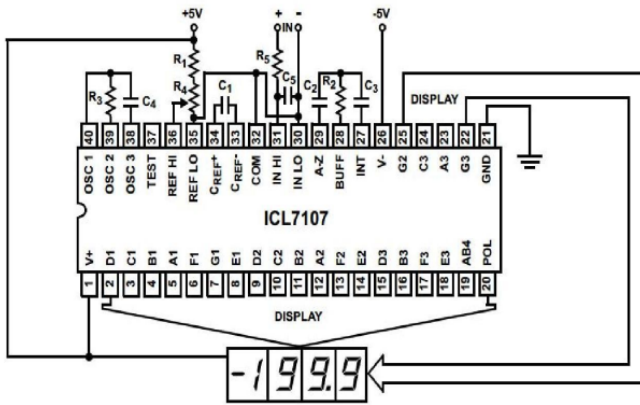


Fig. 6. ICL7107 test circuit and typical application with LED display components, selected for 200 mV full scale.

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